

S. SHERIFRED SINGH

Embedded Software Engineer

Coimbatore, India • +91-7904919946 • ssherifred@gmail.com • [linkedin.com/in/sherifredsing](https://www.linkedin.com/in/sherifredsing)

PROFESSIONAL SUMMARY

- Embedded Software Engineer with over 6 years of overall industry experience, including 4+ years of dedicated embedded software development and 2 years in hardware testing and PCB fault analysis.
- Experienced in Embedded C firmware development for PIC, STM32 and ESP32 microcontrollers with expertise in driver development, hardware-software integration, real-time embedded systems and RTOS-based applications.
- Proficient in UART, SPI, I2C, ADC, Timers, GPIO, Interrupts, PWM, and EEPROM-based embedded applications.
- Strong understanding of embedded architecture, interrupt-driven firmware design, state-machine implementation, and embedded software development lifecycle concepts.
- Strong hands-on background in hardware validation, PCB debugging, hardware-software integration, and debugging complex systems using oscilloscopes and multimeters.

TECHNICAL SKILLS

- **Microcontrollers** : PIC (18F46K22, 16F887), ARM Cortex-M3 (STM32F103), ESP32
- **Programming Languages** : Embedded C, C
- **Communication Protocols** : UART, SPI, I2C, RS-232
- **MCU Peripherals** : ADC, Timers, GPIO, Interrupts, PWM, EEPROM
- **Embedded Development**: Driver Development, Firmware Development, Hardware-Software Integration, State Machine Design, Real-Time Debugging
- **Testing & Validation** : Firmware Testing, Integration Testing, Hardware Validation
- **Embedded Concepts** : Interrupt Handling, Embedded Architecture, SDLC Fundamentals, State Machine Design
- **Sensors & Drivers** : NTC 10K, PT100/MAX31865, MAX6675 (SPI), J-Type TC, GSM SIM800A, DS1307 RTC
- **Development Tools** : MPLAB X IDE, STM32CubeIDE, Git Version Control
- **Hardware Skills** : Hardware Validation, PCB Testing & Debugging, Soldering (SMD & Through-Hole), Oscilloscope, Multimeter, Power Management

PROFESSIONAL EXPERIENCE

Embedded Software Developer

Jan 2022 – Present

NK Automation, Coimbatore

- Developed and maintained **Embedded C firmware** for **PIC18 microcontrollers** used in industrial automation and real-time control systems.
- Designed and implemented **interrupt-driven firmware** with low-level drivers for **UART, SPI, I2C, ADC, Timers, EEPROM, RTC, LCD, and GLCD** peripherals.
- Integrated industrial sensors and external devices including **PT100, MAX31865, MAX6675, DS3231 RTC, and AT24C32 EEPROM** for data acquisition and logging.
- Developed control algorithms for **production monitoring, temperature control, and Automatic Mains Failure (AMF) & Generator Control** applications.
- Optimized firmware for improved **execution speed, memory utilization, and system reliability** through efficient embedded software design.
- Performed **hardware bring-up, hardware-software integration, debugging, and system validation** using oscilloscopes, multimeters, and serial communication tools.
- Collaborated with cross-functional teams to troubleshoot field issues, validate embedded systems, and deliver reliable industrial automation solutions.

Testing Engineer

Jun 2018 – Jun 2020

Akarsh Technologies, Coimbatore

- Performed soldering and desoldering on SMD and through-hole PCBs for production, supporting hardware validation, PCB debugging, fault isolation, and component-level rework.
- Analyzed and resolved PCB faults in textile and industrial automation systems.
- Conducted firmware validation and hardware functionality testing to verify design specifications and identify hardware defects.

TECHNICAL PROJECTS

Advanced Multi-Channel Powerloom Monitoring System—PIC18 MCU, Embedded C, I2C, EEPROM, RTC, GLCD

- Architected an interrupt-driven embedded firmware system for real-time industrial textile production tracking, capturing high-speed machine events with zero data loss.
- Developed dynamic calculation algorithms for critical production metrics (picks, meters, beam balance, and operator pay) with a responsive GLCD-based menu navigation interface.
- Integrated DS3231 RTC and AT24C32 EEPROM via I2C protocol for shift-wise data logging, ensuring 100% seamless data retention and recovery during industrial power failures.
- Executed end-to-end hardware-software integration, system validation, and rigorous field testing in high-noise industrial environments.

Digital Temperature Controller using PT100 + LCD Display—PIC MCU, MAX31865, SPI, Embedded C

- Developed Embedded C firmware interfacing PT100 RTD via MAX31865 SPI driver for precise temperature acquisition.
- Implemented modular LCD driver for real-time display of temperature, setpoint, and system power status.
- Designed relay control algorithm to regulate heater operation based on user-programmable thresholds.
- Optimized system using timer interrupts for stable ADC sampling, debounce handling, and flicker-free LCD updates.

Intelligent Auto Mains Failure (AMF) & Generator Controller —PIC MCU, Embedded C, SPI, Timers, Relays

- Architected interrupt-driven firmware achieving ~98% reliable detection of grid undervoltage and phase imbalance using calibrated ADC sampling with noise-filtered voltage sensing.
- Implemented a deterministic multi-state control algorithm for reliable grid-to-generator switching with safe break-before-make interlocking logic.
- Optimized generator control routines (cranking, re-crank, warm-up, cool-down) with EEPROM-based configuration storage, reducing false crank attempts by ~40%.
- Engineered fail-safe protection layer with relay control, phase validation, and fault-handling logic, reducing unsafe switching by over 60%.

CERTIFICATIONS & TRAINING

Diploma in Embedded Systems

Jun 2020 – Jan 2021

Manfree Technologies, Coimbatore

- Hands-on training in ARM Cortex-M (STM32F103C6), ESP32, and PIC microcontroller programming using C and C++.
- Peripheral interfacing: DS1307 RTC, RS-232, GSM (SIM800A), UART, SPI, I2C, and Bluetooth.
- Tools: Keil, MPLAB, DevC++, and STM32CubeIDE.
- Completed mini-projects in sensor acquisition, communication modules, and embedded control systems.

EDUCATION

Coimbatore Institute of Technology

2021 – 2024

B.E. in Electrical & Electronics Engineering (Part-Time) | Coimbatore, India

Government Polytechnic College, Anna University

2015 – 2018

Diploma in Electronics & Communication Engineering | Coimbatore, India | 72%

Vimal Jyothi Convent Matric Higher Secondary School

2015

Higher Secondary Certificate | Coimbatore, India | 71%